

UNCLASSIFIED



Foundation Sustainment Services (FS2)
Contract HM047621D0009
Aero Task Order PWS
ACE/GTO Sustainment and Modernization
Project Addendum

05 December 2023

Version 4

UNCLASSIFIED

Table of Contents

1	Purpose.....	3
2	Background.....	3
3	Scope.....	4
4	ACE/GTO Current State.....	5
5	High Level Execution Phases	5
6	Services.....	6
7	List of CDRLs	6
8	List of Appendixes.....	8
	Appendix E – ACE/GTO Workflow	9
	Appendix F - Performance Metrics	9
	Appendix G - ACE/GTO Quality Assurance.....	10

1 Purpose

The purpose of this Addendum is to provide O&S services to support the IPF Publications contract to modernize Automated Charting and Encoding (ACE) software for aeronautical Geospatial Terminal Operations (GTO). Additionally, this Addendum provides scope to assume O&S of the existing legacy COE On-Prem system at NDW until cutover to the cloud-hosted system is completed.

This intent is to define the expectations and requirements needed to provide Program Management, Software Maintenance, and Operations Support services as defined in the FS2 Base IDIQ and Aeronautical Task Order, for the ACE/GTO Sustainment and Modernization project. This Addendum establishes the acquisition framework for delivering Operation Support services required for the NGA Source Operations & Management Directorate's Foundation GEOINT Group (SF), Aeronautical Safety Office (SFA) and its mission partners.

The main objective is for the Contractor to design and execute an agile Ops Support construct to deploy and sustain the MDA's legacy as well as a refactored cloud-native solution for ACE/GTO. The modernization end state will provide to the SFA user system performance at a level equal to or exceeding the existing solution, while addressing known and upcoming security, sustainability, and compatibility issues in the legacy code base. The capabilities currently delivered by the legacy ACE/GTO system are to be sustained until directed by the government to be deprecated.

This objective covers the legacy ACE/GTO instances on COE and the future deployment environments of the modernized system identified by SFA and IPF (such as an Unclassified Production Environment).

The FS2 Aero Contractor shall coordinate with the ACE/GTO developer (MDA) to deploy and sustain a refactored ACE/GTO Cloud solution to meet Foundation GEOINT (FG), Aeronautical Safety Office (SFA) production, research and NSG data dissemination requirements.

2 Background

The current ACE/GTO capability production instance is on the NGA COE security domain. The environment is currently UDS data center based under TC APS SPID 61036 and is dated with known long-term sustainability issues that will be addressed over the course of the cloud-modernization effort. This modernization effort will be a partnership between the MDA ACE/GTO COTS development team and the FS2 Aero Team providing Operation Support, integration, A&A and sustainment activities.

The end-state goal is to be accessible and usable from the future production clients that will be offered on the NGA UDS workstation baseline. The future production and dissemination infrastructure of ACE/GTO is therefore likely accessed through Zero-trust thin clients. The FS2

contractor will provide integration guidance as needed for the MDA development team to understand the NGA cloud service environments and options and constraints of the architecture in order to facilitate modernization of the ACE/GTO capabilities.

3 Scope

This Addendum provides the services needed for the sustainment and modernization of MDA Systems' COTS ACE/GTO capability. This includes but is not limited to:

1. Provide operations support needed to migrate the refactored ACE/GTO capability from the current NDW UDS datacenter environment to the designated NGA cloud environment.
2. Provide Ops Support and cybersecurity support for the as-is ACE/GTO capability in the NDW UDS environment.
3. Provide the necessary support to facilitate software SWAP approvals and installation packaging for MDA development releases, in a way that and leverages existing FS2 deployment and operations workflows for test and production environments.
4. Provide test support for software and development environments for ACE/GTO.
5. Leverage NGA GEOINT Services tools and capabilities (e.g. DevSecOps pipelines, OpenShift etc.) to the greatest degree practicable or advantageous to streamline long-term operations and sustainment for the modernized ACE/GTO capability.
6. Provide required software installation and Configuration Management for MDA software releases for both the on-prem and the modernized solution via the FS2 Aero operations & sustainment pipelines.
7. Perform all actions required to ensure that the refactored modernized ACE/GTO solution meets or exceeds all NGA's Risk Management Framework, A&A, Continuous Monitoring to include Cyber security patching requirements within the USCYBERCOM 21-Day requirement (or have an approved POA&M or waiver form).
8. Provision and sustain the virtual infrastructure required for the refactored ACE/GTO software releases in the NGA cloud environment.
9. Provide the cybersecurity, RMF, Assessment & Authorization (A&A), Operations and Sustainment (O&S), and Zero-Trust support services for the refactored ACE/GTO solution needed to progress to Final Operational Capability (FOC).
10. Provide decommission/remove/retire services to shut down the legacy UDS ACE/GTO solution architecture as directed by the government.

4 ACE/GTO Current State

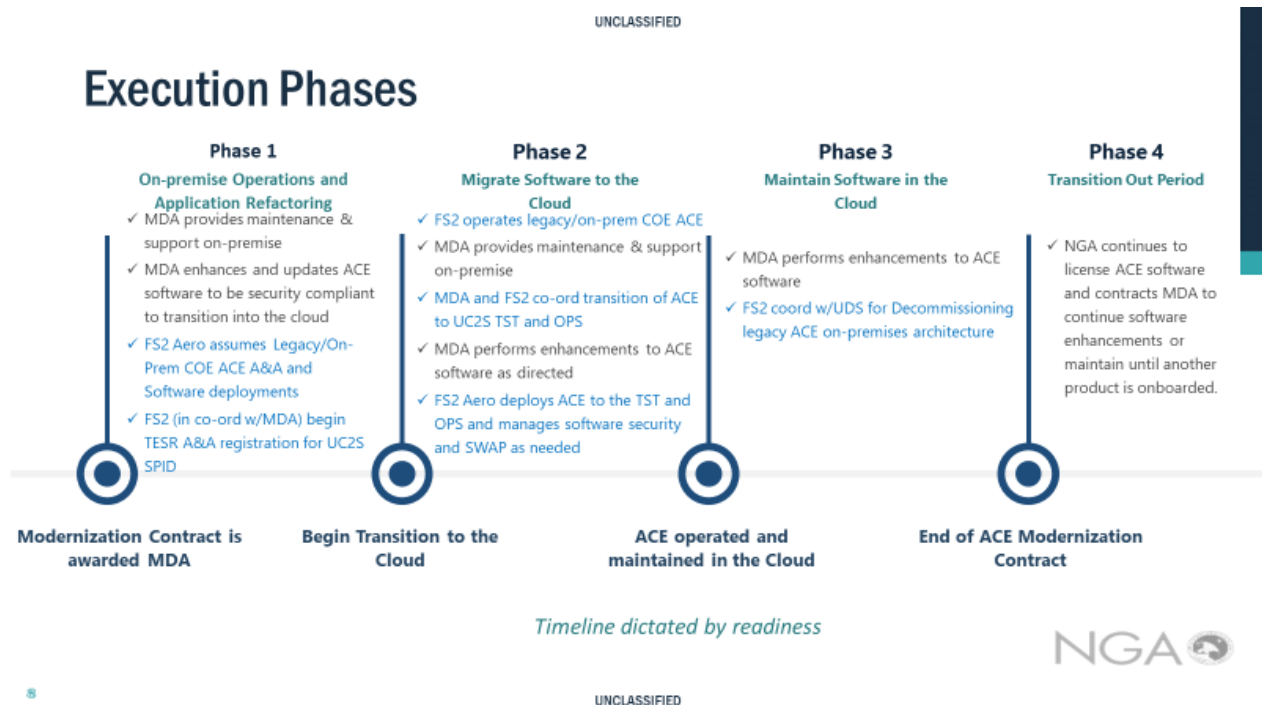
The ACE/GTO system is currently FOC on COE SPID 61036 in UDS environment at NDW. ACE COTS software maintenance updates provided by MDA Systems, are deployed manually by NGA aeronautical personnel.

5 High Level Execution Phases

The Contractor shall establish a software sprint integration and deployment cycle at a schedule/length as jointly agreed to by the Government and the Contractor. This length will be based on coordination dependencies with the MDA Systems development team, the FS2 Aero Team, IPF and SFA government.

At the end of each ACE/GTO Ops Support, integration and Release deployment cycle, the Government and Contractor will evaluate the progress milestones for the ability to meet mission requirements, remediate the liens, and operating cost optimization.

The high-level Phase sequence of the joint FS2 and MDA ACE/GTO Modernization project is depicted below:



5.1 Contract Requirements

The Contractor shall comply with all the Base IDIQ SOW, the IPF Capability Delivery SOP, IPF Software Solution Stack Standards, NGA Software Way, the NGA Technology Strategy, NGA Architecture Design Elements (ADE), and the NGA Web Property Standards and Guidelines (NGAI 8955.6).

6 Services

6.1 Program Management Support (WBS 1.0)

The Contractor shall provide Program Management Services (WBS 1.0) for the modernization effort per the Aeronautical Safety of Navigation Task Order PWS, section 6.1. (CDRL A001)

6.2 Operations Support Services (WBS 5.0)

The Contractor shall provide Program Management Services (WBS 5.0) for the modernization effort per the Aeronautical Safety of Navigation Task Order PWS, section 6.4.

The ACE/GTO capability currently follows a workflow as defined in Appendix E, but may be modified based on the modernization of ACE/GTO per this Addendum, or other Aeronautical-related systems directly affecting this capability.

7 List of CDRLs

This addendum uses the CDRL list from the Base IDIQ and does not have any extra CDRLs unique to this addendum. CDRLs are noted in each section of the WBS

CDRL Title	CDRL #	Data Item Description	Reference	Initial Addendum Award	Subsequent Versions
End of Sprint Status Reports	A004	CFA	6.2	n/a	End of Sprint
Security Plan	A005	CFA	6.2, 6.3	Draft at Start of Phase B	End of each Sprint in which cyber / information security is affected
Transition-In Plan	A006	CFA	6.3	Draft at Start of Phase C	Final by Start of Phase D
Cloud Migration Plan	A009	CFA	6.2, 6.3	Award +60 Calendar Days	As directed by the Government

UNCLASSIFIED

Software Code and Software Code Documentation	A011	CFA	6.2	Award +60 Calendar Days	End of each Sprint
Transition-Out Plan	A012	CFA	6.3	Draft at Start of Phase C	Final by Start of Phase D
Version Description Document	A019	CFA	6.3	Award +60 Calendar Days	End of each Phase
Release Plan	A021	Update JIRA and a PDF copy to PMO	6.3	5 calendar days after first RPE	5 calendar days after each RPE
Value Engineering Change Proposal	A022	CFA	6.2	n/a	As appropriate during Phase A-E
Task Order Kickoff Presentation	A025	Power Point	5.2	At the Kickoff meeting	15 calendar days after future subsystem is added to the Task-Order
Test Readiness Review checklist	A027	CFA	6.2	Deliver to the Government 30 calendar days before the first TRR event	Deliver to the Government 30 calendar days before each TRR event
Operations Support Plan (OSP)	A030	CFA	6.3	Draft at start of Phase C	Final by Start of Phase D
Design Review Presentation (DRP)	A032	CFA	5.2, 6.3	At each Phase Review Milestone	After Phase Review Milestone updated as directed by the PMO
Requirements Traceability Matrix (RTM)	A033	CFA	6.2	At end of Phase A	Updated at end of each Sprint as needed
Capability Integration Schedule (CIS)	A034	CFA	6.2	At end of Phase A	Updated at end of each sprint thereafter

System Performance Optimization Technical Report	A037	CFA	6.3, 6.4	n/a	As directed by the Government
Automation Deployment Instructions	A039	CFA	6.2, 6.3	First Deployment	Every deployment
Operational Readiness Review (ORR) Checklist	A041	CFA	6.2, 6.3	15 calendar days prior to readiness event	Updated after test event as directed by PMO
Integration Test Report	A044	CFA	6.2, 6.3	As directed by Government	As directed by the Government
Corrective Action Report (CAR)	A046	CFA	6.3	As directed by Government	As directed by the Government

8 List of Appendixes

This table is Unclassified

Appendix	Documents
List of Appendixes in the ACE/GTO Addendum	
Appendix E – ACE/GTO Workflow	ACE/GTO Addendum
Appendix F: Performance Metrics	ACE/GTO Addendum
Appendix G: ACE/GTO Quality Assurance	ACE/GTO Addendum
List of Appendixes in Aeronautical Safety of Navigation Task Order PWS	
Appendix A: Performance Metrics	Aeronautical Safety of Navigation Task Order
Appendix B: CDRs	Aeronautical Safety of Navigation Task Order
Appendix C: Technical Library Contents List	Aeronautical Safety of Navigation Task Order

Appendix D: Key Personnel Job Descriptions and Qualifications

Aeronautical Safety of
Navigation Task Order

Table 9-1 – List of Appendices

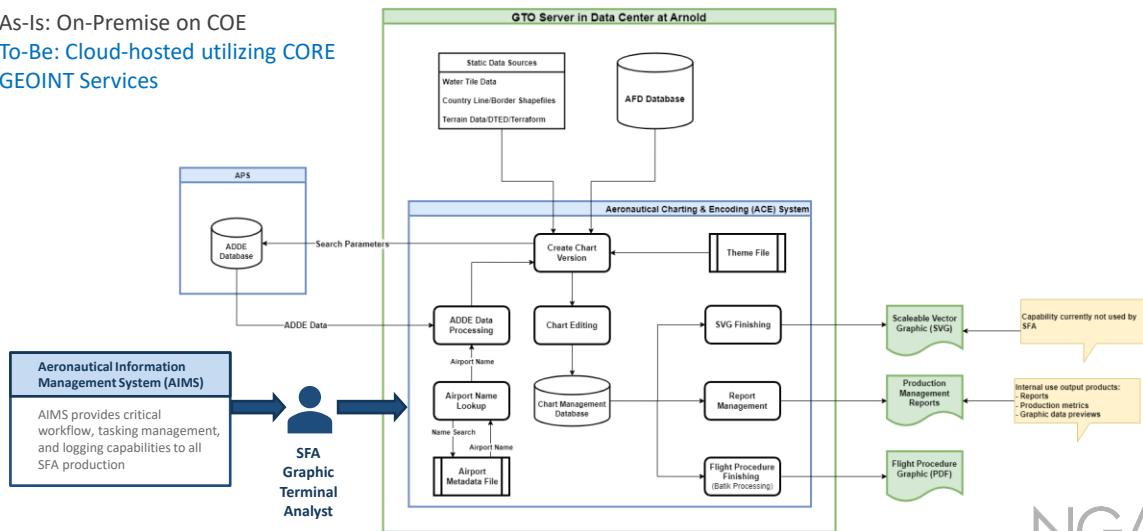
Appendix E – ACE/GTO Workflow

ACE/GTO NDW Implementation

UNCLASSIFIED

ACE Workflow

As-Is: On-Premise on COE

To-Be: Cloud-hosted utilizing CORE
GEOINT Services

NGA

6

UNCLASSIFIED

Appendix F - Performance Metrics

The Contractor shall meet or exceed all the Performance Metrics in this table and the Performance Metrics in the Aero Task-Order PWS (where applicable)

Appendix G - ACE/GTO Quality Assurance

This Addendum will use the Aeronautical Task Order QASP.